

PALLETPILOT · REQUIREMENTS

PalletPilot requirements

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DOCUMENT CONTROL

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Project TRL	TRL 2 Technology concept formulated		
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REVISION HISTORY

Version	Date	Author	Change
0.2	2026-09-25	Amish Chadha	First measurable requirements for TRL 2, with status against the concept estimates
0.1	2026-09-24	Amish Chadha	Initial scaffold

PalletPilot requirements

These are first-pass requirements for the concept. Targets are proposals for review, not yet validated with users, and will be checked by calculation at TRL 3 and revised after site visits (see PLP-PRB-001). The status column compares each target with the first-order estimates in PLP-PRC-001; every status is an estimate until verified.

The **design load case** is a 1,000 kg pallet on a 64 kg donor jack with a kit of about 37 kg, about 1,100 kg in total, on a level, dry, sealed concrete floor.

The **design duty** is one 8 h shift of 60 pallet moves, each 40 m loaded and 40 m empty, with four starts in each direction.

Table 1. Requirements.

ID	Requirement	Target	Verification (TRL 3 or later)	Status (estimate)
R1	Fit a common manual jack	Bolts to 27 x 48 in (685 x 1220 mm), 2,500 kg manual jacks with a common steering yoke; no welding, cutting or drilling of load-bearing parts; fitted in 2 h or less and removed in 1 h or less with hand tools	Survey of three donor models; fitting sequence review	Unverified; yoke patterns vary
R2	Move the design load	Start and move 1,000 kg on a level floor; kit rating 1,500 kg at reduced speed (0.8 m/s or less), below the donor rating	Traction and torque calculation	Met on paper at 1,000 kg; traction margin thin at 1,500 kg
R3	Handle short ramps	Start, climb and hold the design load on a 2 % grade	Traction, torque and brake calculation	At risk: needs about 350 N of 550 N available traction
R4	Walking-pace speeds	Walk mode 1.2 m/s (4.3 km/h) or less handle-end first and 0.8 m/s or less forks first; creep 0.3 m/s or less with the handle near upright; follow mode 0.6 m/s or less	Controller limits; later timed runs	Met by design (software limits)
R5	Follow the operator	In follow mode, hold a 1.5 m gap within ± 0.3 m and a bearing within $\pm 10^\circ$ of the tag at walking pace	UWB error budget; later tracking trials	Not met on paper: raw bearing error about $\pm 18^\circ$ with a 450 mm anchor baseline; needs filtering or angle-of-arrival sensing
R6	Stop if the operator is lost	Stop within 0.3 s of losing the tag link, a gap over 3 m, a tag stop press or the tag leaving a $\pm 45^\circ$ cone	Firmware design review; later bench test	Unverified
R7	Stop before hitting a person in follow mode	Detect a person in the travel path and stop without contact from 0.6 m/s with the design load	Stopping calculation; later test with an ISO 3691-4 style test piece	Not met: a contact bumper alone cannot stop 1,100 kg from 0.6 m/s within its travel (about 0.42 m stop versus about 40 mm travel); needs non-contact sensing or a lower speed
R8	Stop on bumper contact	Bumper contact removes drive power and applies the brakes within 100 ms; bumper-only stopping limited to 0.2 m/s or less	Stop chain design review	Met by design at 0.2 m/s (about 0.06 m stop)
R9	Emergency stop	Two hardwired emergency stops (tiller head and enclosure) plus the tag stop; the hardwired stops cut drive power through a safety relay and contactor, independent of software; target performance level PL d under ISO 13849-1	Stop chain review; PL calculation at TRL 3	Unverified
R10	Park safely	Brakes apply with power off and hold the design load on a 2 % grade	Brake torque check against datasheet	Unverified; needs about 22 N·m holding torque in total
R11		Walk mode drives only with the handle between about 20° and 70° from vertical;	Design review	Met by design

ID	Requirement	Target	Verification (TRL 3 or later)	Status (estimate)
	Keep the operator in control	belly-reverse paddle on the tiller head reverses the truck when pressed		
R12	Work a full shift	Complete the design duty on one charge with 20 % or more of usable energy left	Energy calculation	Met: about 281 Wh used of about 410 Wh usable (31 % left)
R13	Charge overnight	Full charge in 5 h or less from a 120 V or 230 V outlet through a certified charger	Charger datasheet	Met: about 4.5 h at 5 A
R14	Push by hand when unpowered	One lever lifts the drive wheels clear and releases the brakes in 10 s or less; unpowered push force no more than 10 % above the bare jack	Design review; later push-force test	Unverified
R15	Keep the jack usable	Kit mass 40 kg or less; overall length increase 300 mm or less; lowered fork height and pallet entry unchanged	Mass estimate; model check	Partly met: about 37 kg, but about 330 mm longer at floor level with the bumper (about 100 mm beyond the upright handle)
R16	Indoor environment	Operate at 0 to 40 °C; electronics and connectors IP54; no charging below 0 °C	Datasheets and design review	Unverified
R17	Affordable	Kit parts cost \$1,200 or less, donor jack excluded	Priced BOM (bom/ bom.csv)	Not met: about \$1,425, about 19 % over

■ Requirements not met or at risk

- **R7, non-contact stopping in follow mode, is not met.** This is the main finding of the concept. See PLP-PRC-001, "Stopping".
- **R5, follow bearing accuracy, is not met on paper** with two range-only anchors; it may be met with filtering or angle-of-arrival modules.
- **R17, cost, is not met** by about \$225. A walk-only variant costs about \$1,075.
- **R3, the 2 % ramp, is at risk** on traction margin.
- **R15 is partly met.** Mass is within target, but the drive module and bumper add about 330 mm at floor level, 30 mm over the target, which widens the turning circle in tight aisles.

■ Assumptions

- Rolling resistance 1.2 % and breakaway resistance 2.5 % for polyurethane steer and load wheels on smooth concrete. These are assumptions to be measured on a real jack.
- Tread-to-floor friction coefficient 0.5 for the drive wheels on dusty concrete; lower on wet or polished floors.
- The drive wheels carry about 1.1 kN of spring preload and no share of the pallet load.
- Motor plus driver efficiency 70 %; controls, UWB and lights draw about 8 W in total.
- Operator reaction and stop-chain delay of 0.1 s in follow mode.
- The legal status of the retrofit (OSHA 1910.178(a)(4)) and of follow mode (ISO 3691-4) is open; targets R7 and R9 anticipate the stricter answer.